AAE 552—Nondestructive Evaluation of Structures and Materials
3 credits

1 Course Information/Description

Overview the physics, principles, and methods employed for nondestructive evaluation (NDE) of structures and materials. Scope includes both practical considerations and underlying physics. Major NDE techniques covered include radiographs, ultrasonics, eddy currents, penetrants, magnetic particle, thermography, and visual/optical methods.

2 Learning Resources and Reference Text

Brightspace Learning Management System:

You can access the course via Brightspace. It is strongly suggested that you explore and become familiar not only with the site navigation, but with content and resources available for this course. See the Help tab for resources. Course notes, homework, study guides, exams, and lecture videos will be posted on Brightspace.

Textbook:

Fundamentals of Structural Integrity by Alten F. Grandt, Jr.

3 Assignments

3.1 Homework

Homework sets will be assigned corresponding to each major topic in the course. Homework is collectively worth 35% of your overall grade. Late homework will not be accepted. Working with classmates is encouraged, but simply copying answers will be considered cheating and receive no credit. The use of AI or LLMs is forbidden and is punishable by failing the assignment. The grader may use his or her discretion in the aforementioned.

3.2 Term paper and term paper sub-assignments

A term paper examining a case study, industrial practice, or research-based NDE is due at the end of the semester and worth 25% of your grade. Supporting documents leading up to the term paper are worth an additional 15% of your grade. The term paper must follow templates provided on Brightspace. Grading of the term paper will be discussed in class. The use of AI or LLMs is forbidden and is punishable by failing the sub-assignment or the entire term paper.

4 Course Learning Objectives

The objective of this course is to give students an understanding of the importance of NDE, the underlying physics of common NDE techniques, and the processes by which these techniques are applied.

5 Evaluation

Grading is weighted as shown below. A breakdown of the grade scale is also provided below. The instructor reserves the right to curve or to not curve the class.

Grading weights:

- 5% - half-page proposal of term paper topic
- 5% - list of 6 references and relevance to term paper
- 5% - outline of term paper
- 35% - homework
- 25% - term paper
- 25% - final exam

Grade scale:

- A+ ≥ 96.67% (both an A+ and an A are worth 4.0)
- 96.67% > A ≥ 93.33%
- 93.33% > A- ≥ 90%

- 90% > B+ ≥ 86.67%
- 86.67% > B ≥ 83.33%
- 83.33% > B- ≥ 80%

- 80% > C+ ≥ 76.67%
- 76.67% > C ≥ 73.33%
- 73.33% > C- ≥ 70%

- 70% > D+ ≥ 67.67%
- 67.67% > D ≥ 63.33%
- 63.33% > D- ≥ 60%

- F < 60%